

ASUMBI GIRLS' 2025 PREDICTION



CONFIDENTIAL KCSE EXAM



443/1

AGRICULTURE

PAPER 1

TIME: 2 HOURS

NAME.....

SCHOOL.....

SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

Instructions to students

- a) Write your name and Admission number in the spaces provided at the top of this page*
- b) This paper consists of three sections: A , B and C*
- c) Answer all questions in sections A and B*
- d) Answer any two questions in section C*
- e) All answers should be written in the spaces provided*
- f) Candidate should answer the questions in English*

FOR EXAMINER'S USE ONLY

SECTION	QUESTIONS	MAXIMUM SCORE	CANDIDATE SCORE
A	1-21	30	
B	22-25	20	
C		20	
		20	
	TOTAL SCORE	90	

SECTION A (30 MARKS)

Answer all questions in this section on the spaces provided

1. Suggest **three factors that determine the choice of a method of farming**

(1½ marks)

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2. Outline **three effects of soil organisms which benefit crop growth**

(1½ marks)

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3. State the soil component that determines

(1mark)

Soil Structure class

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Soil Structure grade

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4. State **four effects of hard pans in the soil**

(2 marks)

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5. State **four factors that determine the number of secondary operations carried out on**

land in the

formation of a seedbed

(2marks)

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6. State **four** factors that are hinder access of enough and clean water to most Kenyans communities (2marks)

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7. Name **three** causes of water logging in arable land (1½ marks)

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8. What is the difference between compost manure prepared in drier areas and one prepared in wet areas? (1 mark)

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9. State **two** methods used to test the PH of the soil (1 mark)

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10. Differentiate between stem and root tubers (1 mark)

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11. The average spacing of Arabica coffee is 2.7 m by 2.7 m. Calculate the plant population in one hectare (1Ha) of land (*Show your working*) (2marks)

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12. State any **two** practices carried out during hardening of seedlings in the production of tomatoes (1mark)

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13. Name the **three** types of suckers that are supposed to be left in each banana stool after removal or pruning of the extra suckers. (1½ marks)

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14. State **four** types of information contained in a land title deed (2 marks)

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15. Explain **two** ways in which grass strips help to conserve the soil (1 mark)

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16. State **four** disadvantages of tillage as a method of weed control (2 marks)

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17. Explain the difference between pests economic injury level and integrated pest management (1mark)

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18. Name the **three** grades of coffee obtained after sorting out the picked berries (1½ marks)

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19. What is topping in pasture management (½ mark)

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20. State **three** ways in which labour productivity can be improved (1½ marks)

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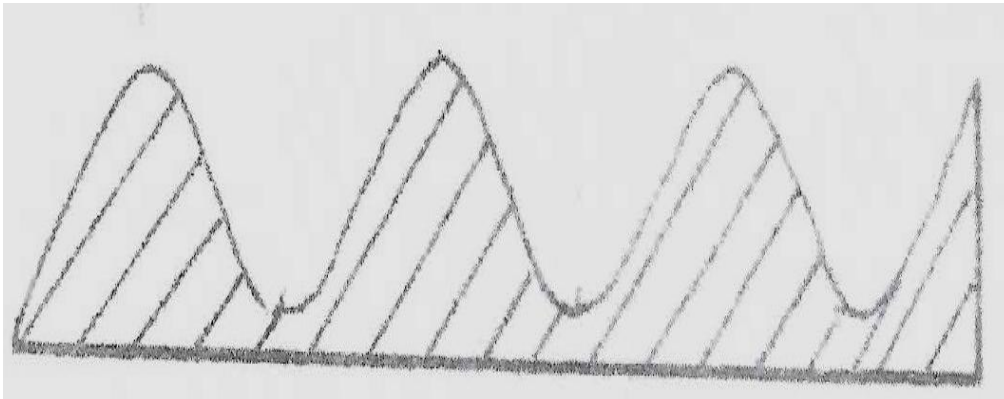
21. State **three** roles of agricultural co-operatives in Kenya (1½ marks)

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SECTION B (20 Marks)

Answer ALL questions in this section on the spaces provided

22. Below is a diagram of a tertiary land operation. Study it and answer the questions that follow



a) Name the tertiary land operation

(1 mark)

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b) Explain the process of performing the tertiary land operation

(1 mark)

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c) Name the tractor implement that can be used to achieve the operation

(1

mark)

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d) State the importance of carrying out the tertiary operation

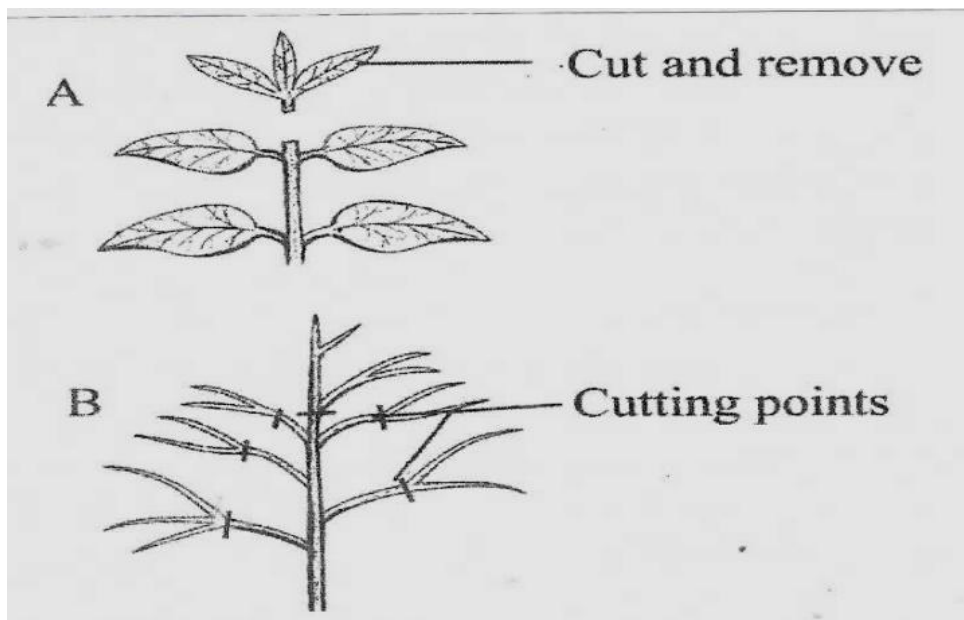
(2 marks)

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23. Below are diagrams of two methods of pruning. Study them carefully and answer the questions that follow.



i. Identify the pruning methods shown below

(2 marks)

Method

A

Method

B

ii. When would it be necessary to carry out the type of pruning illustrated in both cases(2 marks)

Method

A

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Method

B

.....

iii. Give **two** advantages of carrying method of pruning in tomatoes

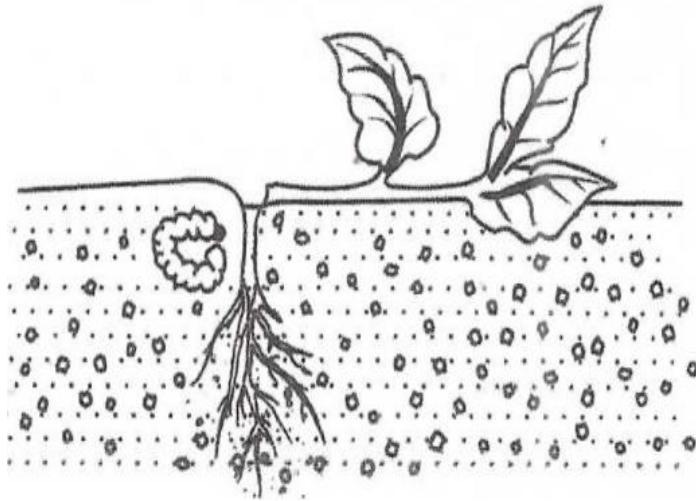
(1mark)

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24. Below is a diagram of a cabbage pest. Study it carefully and answer the questions that follow



a) Identify the cabbage pest shown above

(1 mark)

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b) In which category of pests does the pest you named in (a) above belong

(1mark)

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c) What kind of mouth part does the pest named in (a) above possess?

(1 mark)

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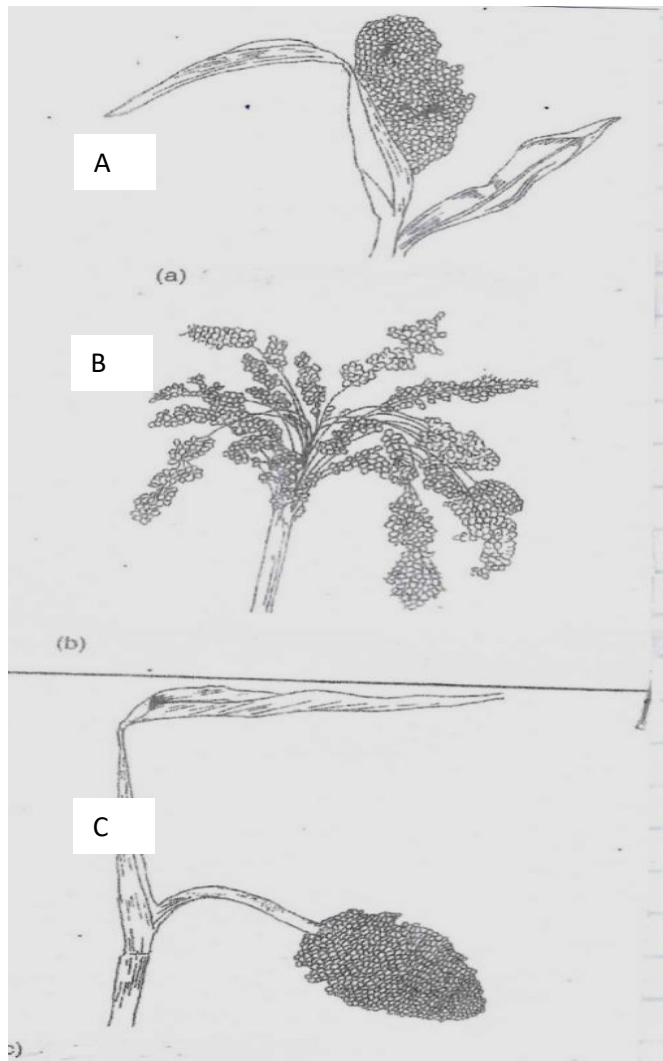
d) Suggest **two** control measures for the pest named in (a) above

(2 marks)

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25. Below are diagrams of different varieties of sorghum. Study them and answer the questions that follow



i. Identify the different varieties

(3 marks)

A.....
B.....
C.....

ii. State **one** advantage of the variety labeled C over the other varieties

(1 mark)

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iii. Name **two** common pests of sorghum

(1 mark)

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SECTION C (40 Marks)

Answer any TWO questions in this section in the spaces provided after question 28.

26. a) State and explain **four** ways in which biological agents may enhance soil formation
(6 mks)

b) Discuss **six** reasons on the importance of keeping good and up to date farm accounts in the farm

(6 marks)

c) Discuss why liming is a recommended practice in crop production (4 marks)

d) Explain the roles of irrigation water in rice production (4 marks)

27. a) Outline the procedure undertaken in inheriting a registered land upon death of the owner

(8 marks)

b) State the **three** harvesting practices of maize (3 marks)

c) Discuss the processes of large scale chemical treatment of water (9 marks)

28. The following information was obtained from Keheo farm. Study the information and prepare a balance sheet for the farm for the year ended 31st JULY/AUG 2019

Payable tax	-	16,000
Interest on loan	-	24,000
Payable debts	-	14,000
Two year bank loan	-	120,000
Wages	-	24,000
Buildings	-	84,000
Receivable debts	-	12,000
Livestock	-	124,000
Feeds in store	-	12,000
Maize in store	-	42,000
Cash at hand	-	22,000
Movers	-	5,500
Cash in bank	-	52,000

7 marks

ii) Was Keheo farm solvent or insolvent? (1 mark)

b) State **six** advantages of mixed cropping (6 marks)

c) Name the **three** zones of the production function and explain what happens in each of the

production zone

(6 marks)